Problem Chosen	2025	Team Control Number
C	MCM/ICM Summary Sheet	2515734

Research on the prediction of National Olympic medals and the "great coach" effect

The Olympic medal tally is a global focal point, reflecting the sporting strength, investment, and development of countries. Predicting Olympic medals and evaluating the "great coach effect" can help national Olympic committees formulate better preparation strategies, identify potential sports and athletes, and reveal performance trends for future Olympics.

In the first question, to address the issue of the dispersion of athletes' medal counts, we adopted a multivariate negative binomial model to predict the medal counts and the medal achievements of various countries for the 2028 Los Angeles Olympic Games. We also performed uncertainty analysis on these predicted medal counts, deriving the corresponding prediction intervals. Subsequently, we used a Bayesian hierarchical model to predict the countries that are likely to win medals for the first time. This prediction model, by incorporating a hierarchical structure, enhances prediction accuracy by facilitating information exchange and sharing across different data levels, even in the presence of incomplete data. Finally, to better predict the medal performance of new events, we selected existing events highly similar to the new ones based on event characteristics and historical data similarities, and performed medal predictions using a Bayesian model. According to the results, athletics and swimming made the most significant contribution to predicting gold medals, followed by wrestling, gymnastics, and shooting. These events had even more pronounced effects on the predictions for silver and bronze medals, showcasing the strength of countries in these traditional strong sports. Additionally, the host country played a significant role in boosting the predicted medal counts.

For the second problem, the author uses the hypothesis test method to explore the teaching effect of Liu Guoliang, Li Yongbo and Li, and reveals the significant influence of Liu Guoliang, Li Yongbo and Li Falu on the number of medals in the sports team. Through the comparative analysis, we quantify the positive changes brought by these coaches during the teaching period by using the paired sample t-test and regression analysis. The results show that the number of medals obtained by the teams under their guidance in the corresponding matches has significantly increased, and the average score of the medals has increased by 9 points, 7.6 points and 5.2 points respectively. This finding further confirms the existence of the "great coach" effect. In order to make better use of this coach effect, this article also provides India, Denmark and Japan with reasonable coach investment suggestions through the prediction of the return on coach investment.

For question three, our model reveals several original insights, including similarities in the performance of medals among countries, host nation effects, the influence of individual athlete differences on medals, quantification of coach effects, and the predictive potential of external data such as social media. These insights provide the basis for IOC decisions to optimize readiness strategies, resource allocation, athlete selection, and coach investment, thereby helping them improve medals performance and plan more scientifically.

Keywords: Prediction,Multivariate negative binomial, Bayesian Hierarchical Model,t-test, Regression Analysis.

Contents

1 Introduction

1.1 Background

The Olympic Games, as a grand sporting celebration, offer more than just the fierce competition among athletes; the Olympic medal tally also captivates the attention of global audiences. The competition among nations on the medal tally not only demonstrates their athletic prowess but also reflects their investment and progress in sports.

Therefore, predicting Olympic medals holds multiple significances. It provides more scientific preparation guidance for national Olympic committees, assisting countries in identifying potential advantageous events and athletes. Furthermore, accurate predictions can reveal the development trends of various countries in future Olympic Games, providing a solid basis for the formulation of sports policies and investment decisions. By constructing reasonable prediction models, we can not only gain insightful foresight into the medal tally of the 2028 Los Angeles Olympic Games but also explore ways to further enhance the overall sports competitiveness of nations through factors such as the guidance of outstanding coaches and the selection of event categories.

Figure 1: Olympic medals

1.2 Problem Restatement

Predicting Olympic medals for various countries presents a multifaceted challenge that necessitates consideration of numerous factors. Based on the problem background and conditions provided in the article, our focus will be directed towards the following critical issues within the context of mathematical modeling:

1. **Development of a Model for Medal Prediction:** This involves constructing a model capable of predicting medals counts for different countries, inclusive of estimating the model's uncertainty/accuracy and assessing its performance.

2. **Application of the Model to the 2028 Los Angeles Olympics:** Utilize the model to provide prediction intervals for all outcomes, analyze which countries are likely to make progress compared to their performance at the 2024 Paris Olympics, and identify those whose performance may decline.

3. **Prediction of Countries Potentially Winning Their First Medal at the 2028 Los Angeles Olympics:** Estimate the likelihood of specific countries achieving this milestone and quantify the results.

4. **Exploration of the Relationship Between Medal Counts and Event Categories:** Investigate how different events contribute to medal tallys for various countries. Which events are more significant for different nations and why? How do the events chosen by home countries influence competition outcomes?

5. **Estimation of the "Great Coach" Effect on Medal Counts:** Based on the definition of a "great coach," quantify their contribution to medal tallies. Select three countries and determine the sports in which they should consider investing in great coaches, estimating the potential impact.

6. **Identification of Additional Original Insights from Model Analysis:** Extract further insights from the model analysis related to Olympic metrics, explaining how these insights can inform national Olympic committees.

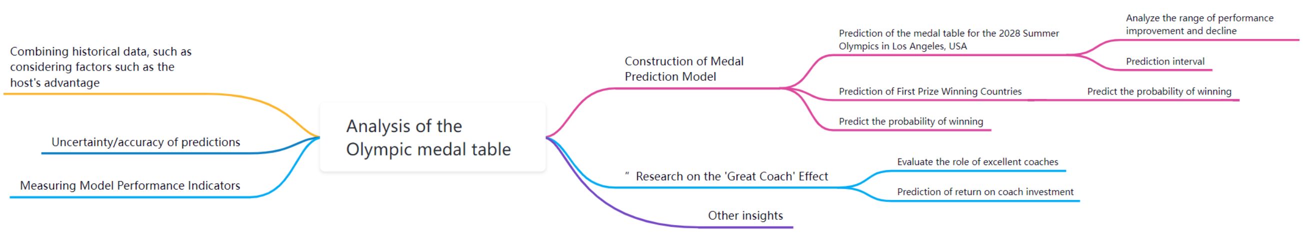


Figure 2: Process diagram for solving problems

2 Assumptions and Justifications

Our model establishment will be based on several assumptions, which we will elaborate on here:

- **Assumption 1:** The medal counts in different Olympic events are independent.
  Justification: Different competitions have varying physical demands, skill requirements, and completely different rules. Therefore, our assumption is reasonable.

- **Assumption 2:** An athlete's probability of winning an award in one competition does not affect their probability of winning an award in another competition.
  Justification: Here, we ignore subjective factors such as the athlete's psychological state affecting their performance in competitions. This is necessary for the establishment of our model.

- **Assumption 3:**Athletes do not change their nationality and continue to compete on behalf of their original country.
  Justification: In most cases, a person's nationality is long-term and stable, and not easily changed. Athletes, as individuals, share this stability in their nationality identity.

- **Assumption 4:** Different disciplines are independent of each other.
  Justification: Our model focuses purely on the sport or discipline level, and the total medal count is simply the sum of medals from all events. Therefore, there is no need to make predictions for the overall medal count in a formalized manner.

3 Notations

Symbol	Instructions
i	the ith country
j	the jth year
k	the $k$the kth sport event indicator.
$\mu_{i,j,k}$	The expected count for each type of medal (gold, silver, bronze) for country i, year j, and discipline k.
$\mathbf{x}_{i,j,k}$	A vector of covariates for country i, year j, and discipline k.
μ_{pre}	the mean number of medals won before coaching.
μ_{post}	the mean number of medals won after coaching.
d	the difference between the sample means
s_d	the standard deviation of the differences
d_i	he difference between the number of medals won before and after coaching.
Medal_score_i	the number of medals won in year i
β_0	the intercept term.
β_1	the regression coefficient indicating the effect of coaching change on the number of medals.
ϵ_i	the error term.

4 Task 1:Establishing a Medal Prediction Model

4.1 Data Processing

To ensure model stability, accurate and continuous data is essential. The data in the attachment contained issues such as incorrect values, missing data, and improper punctuation, which could affect accuracy and lead to unstable predictions. Thus, we cleaned and calibrated the data to ensure completeness and consistency, enhancing the model's practical utility and real-world applicability. The raw data was processed as follows:

1. **Deletion of Incorrect Data.** In the raw data, there were errors such as non-official competitions represented by "?" that interfered with our model building. To facilitate data processing, we deleted these data points.

2. **Filling Missing Values.** Some cells contained no data, and we filled these empty cells with "0".

3. **Data Inquiry.** To predict the number of medals for the 2028 Los Angeles Olympics, we needed to know the events scheduled for the games. Therefore, we queried data from the following two websites:

    - General Administration of Sport of China: `https://www.sport.gov.cn/n20001280/n20745751/c2656761`
    - LA28 Website: `https://la28.org/en/newsroom/LA28_New_Olympic_Sports_Proposed.html` (in tabular form)

4. **Data Rearrangement.** To facilitate the counting of national medal totals, we rearranged the medals awarded to countries in `summerOly_medal_counts.csv` in descending order by country and event.

5. **Data Classification and Screening.** Based on the data obtained in Step 3, we added new columns titled "Sport Count" and "Host".

4.2 Medal Count Prediction for the 2028 Los Angeles Olympics Based on a Multivariate Negative Binomial Model

The number of medals won by athletes, as a type of count data, often exhibits overdispersion, which makes the use of a simple Poisson distribution insufficient for modeling. The Poisson distribution assumes that the event rate is constant; however, in reality, the number of medals won by athletes is influenced by a variety of factors, causing the level of dispersion to exceed the applicability of the Poisson distribution. Therefore, a multivariate negative binomial model, which introduces new parameters, can better capture this overdispersion phenomenon. The multivariate negative binomial model not only accounts for heterogeneity among different athletes but also effectively predicts the distribution of medal counts.

Response Variables. Let the response vector be:

$$\mathbf{y}_{i,j,k} = \begin{bmatrix} \text{Gold}_{i,j,k} \\ \text{Silver}_{i,j,k} \\ \text{Bronze}_{i,j,k} \end{bmatrix},$$

where:

- i indexes countries (National Olympic Committees, NOCs),
- j indexes years,
- k indexes disciplines.

Predictors. The predictor variables include:

- Total Participants$_{i,j,k}$: The total number of participants from country i in discipline k for year j,
- Total Events$_{j,k}$: The total number of events held in discipline k during year j,
- Host Country Indicator$_j$: A binary indicator (0/1) for whether country i is the host nation in year j,
- $\mathbf{y}_{i,j-1,k}$: The number of medals (*gold*, *silver*, *bronze*) earned by country i in discipline k during the previous Olympics,
- Interactions between country and discipline, as well as year-discipline effects.

Model Specification. The expected count for each medal type is modeled using a log-link function:

$$\log(\mu_{i,j,k}) = \mathbf{x}_{i,j,k}^{\top}\boldsymbol{\beta} + u_i + v_{j,k} + w_k + \theta_{i,k} + \delta_j,$$

where:

- $\mu_{i,j,k}$: The expected number of medals (*gold*, *silver*, or *bronze*) for country i, year j, and discipline k,
- $\mathbf{x}_{i,j,k}$: A vector of covariates for country i, year j, and discipline k,
- $\boldsymbol{\beta}$: A vector of regression coefficients for the covariates,
- $u_i \sim \mathcal{N}(0, \sigma_u^2)$: Country-specific random effects capturing baseline differences between countries,
- $v_{j,k} \sim \mathcal{N}(0, \sigma_v^2)$: Year-discipline interaction effects capturing variability across disciplines over time,
- $w_k \sim \mathcal{N}(0, \sigma_w^2)$: Discipline-specific random effects,
- $\theta_{i,k} \sim \mathcal{N}(0, \sigma_\theta^2)$: Country-discipline interaction effects capturing specific strengths or weaknesses of countries in particular disciplines,
- $\delta_j \sim \mathcal{N}(0, \sigma_\delta^2)$: Host country effects for year j, capturing the advantage experienced by the host nation.

Response Distribution. The observed medal counts are modeled as:

$$\mathbf{y}_{i,j,k} \sim \text{NegBin}(\mu_{i,j,k}, \phi),$$

where:

- $\text{NegBin}(\mu, \phi)$ denotes the negative binomial distribution with mean μ and dispersion parameter ϕ,
- ϕ: Controls overdispersion in the count data.

The negative binomial distribution is particularly suitable for modeling count data with overdispersion, where the variance exceeds the mean.

Prior Information. Priors are specified for all parameters to regularize the model and reflect domain knowledge:

$$\boldsymbol{\beta} \sim \mathcal{N}(0, \sigma_\beta^2), \quad \sigma_u, \sigma_v, \sigma_w, \sigma_\theta, \sigma_\delta \sim \text{Half-Cauchy}(0, 1).$$

Including Countries with No Prior Medals. For countries that have not earned medals in previous Olympic games, we include them in the model by assigning their prior medal counts ($\mathbf{y}_{i,j-1,k}$) to zero. This allows the model to estimate their medal probabilities based on other covariates such as participation levels, host country status, and discipline-level trends.

Posterior Inference. Posterior inference is performed using Markov Chain Monte Carlo (MCMC) sampling. The posterior distribution of the parameters is given by:

$$p(\boldsymbol{\beta}, u, v, w, \theta, \delta \mid \mathbf{y}) \propto p(\mathbf{y} \mid \boldsymbol{\beta}, u, v, w, \theta, \delta) \cdot p(\boldsymbol{\beta}) \cdot p(u) \cdot p(v) \cdot p(w) \cdot p(\theta) \cdot p(\delta).$$

Prediction for 2028. To predict medal counts for the 2028 Olympics:

1. Compute $\mu_{i,2028,k}$ for each country i and discipline k using the posterior samples of the parameters:

$$\mu_{i,2028,k} = \exp\left(\mathbf{x}_{i,2028,k}^\top \boldsymbol{\beta} + u_i + v_{2028,k} + w_k + \theta_{i,k} + \delta_{2028}\right).$$

2. Simulate medal counts from the predictive posterior distribution:

$$\mathbf{y}_{i,2028,k} \sim \text{NegBin}(\mu_{i,2028,k}, \phi).$$

3. Summarize the posterior predictive distribution to compute point estimates (e.g., posterior means) and credible intervals for the medal counts.

Due to the limited space, each table below only shows part of the data

Table 1: Prediction and confidence interval of National Olympic medals

NOC	Gold	Silver	Bronze	Gold CI	Silver CI	Bronze CI
United States	39	41	40	(36,40)	(36,45)	(37,43)
China	40	27	24	(37,40)	(24,27)	(21,24)
Japan	20	13	12	(14,20)	(11,14)	(9,12)
Australia	18	19	12	(12,19)	(8,19)	(12,17)
France	15	25	20	(12,18)	(18,27)	(16,22)
Netherlands	14	7	13	(10,15)	(6,8)	(9,13)
Great Britain	14	20	25	(11,17)	(18,22)	(22,28)
Italy	12	14	16	(10,13)	(10,15)	(12,17)
Canada	6	7	5	(3,7)	(3,9)	(2,7)
Spain	10	7	5	(7,12)	(5,7)	(4,6)

Table 2: Countries with improved Olympic performance

NOC	2024 Total	2028 Total	Improvement
Saint Lucia	2	7	5
Fiji	1	2	1
Spain	18	23	5
Jordan	1	5	4
United States	126	134	8
Kosovo	0	3	3
Malaysia	2	5	3
Botswana	2	4	2
Kyrgyzstan	2	4	2
Mongolia	1	3	2

Table 3: Countries with declining Olympic performance

NOC	2024 Total	2028 Total	Improvement
South Korea	32.	13	-19
Germany	33	16	-17
France	64	51	-13
Australia	53	43	-10
Canada	27	20	-7
China	91	85	-6
Kenya	11	10	-1
Great Britain	65	59	-6
Iran	10	7	-3
Netherlands	34	30	-4

4.3 Prediction of First-Time Medal-Winning Countries Based on Bayesian Hierarchical Models

Bayesian hierarchical models capture structured information in data by introducing distributions for parameters at different levels and establishing dependencies between these levels. The core idea is to share information through the establishment of a hierarchical structure, enabling the model to have better predictive capabilities when faced with incomplete or noisy data.

4.3.1 Overview of the Bayesian Model Structure:

- Data Layer (Observation Layer):This layer sits at the bottom, responsible for describing the generation mechanism of observed data. Each set of observed data i originates from a specific distribution, which is based on the unique parameters $\theta_i$for each set:

$$y_i \sim P(y_i \mid \theta_i)$$

Here,y_i represents the observed values for the i-th data set.

- Group Layer:Each group (or subset) in the data layer is equipped with its own set of parameters θ_i, which may share commonalities among them. The group parameters often stem from a higher-level distribution. For instance, we can assume that the mean $\mu_i$for each group follows a normal distribution, while the variance σ_i follows a gamma distribution:

$$\theta_i \sim P(\theta_i \mid \mu, \sigma^2)$$

- Hyperparameter Layer:At this level, we define hyperparameters that govern the distribution of group parameters. These hyperparameters themselves are also inferred from prior distributions. For example, we might assume they follow a normal distribution or another appropriate distribution:

$$\mu \sim P(\mu \mid \alpha, \beta), \quad \sigma^2 \sim P(\sigma^2 \mid \gamma, \delta)$$

Here, α, β, γ,δ and are the hyperparameters that define the prior distributions for μ and σ^2 , respectively.

1. Establish Prior Beliefs: In this stage, we meticulously set prior distributions at various levels of the model, including the data level, group level, and hyperparameter level. These prior distributions essentially reflect our initial understanding and beliefs about the model parameters before observing the data. They provide us with a starting point to commence the entire prediction process.

2. Data Observation and Likelihood Construction: Next, we focus on data observation and the construction of the likelihood function. Specifically, each data pointy_i is associated with a corresponding parameter θ_i through a particular distribution (e.g., a normal distribution). This relationship is clarified by defining the likelihood function$P(y_i|\theta_i)$, which describes the mechanism or likelihood of generating the data y_i given the parameters θ_i, This step serves as a crucial bridge connecting the observed data with the model parameters.

3. Derivation of the Posterior Distribution: Finally, we utilize Bayes' theorem to update our beliefs, namely the prior distributions. Combining the actual observations of the data, we compute the posterior distribution of the model parameters:

$$P(\theta_i|y_i) = \frac{P(y_i|\theta_i)P(\theta_i)}{P(y_i)}$$

This process takes into account not only the prior information but also fully incorporates the evidence provided by the data, thereby providing us with a more accurate and comprehensive parameter estimation. The posterior distribution not only reflects our current understanding of the parameters but also lays a solid foundation for our future predictions.$P(y_i)$represents the marginal likelihood of the observed data, which is often not straightforward to compute directly, especially in hierarchical models. Numerical methods (such as MCMC) are required for inference in such cases.

4. Prediction and Inference: Based on the computed posterior distribution, predictions are made for new data. This process typically involves inference at both the hyperparameter level and the group level, and involves sampling from the posterior distribution to generate prediction results.

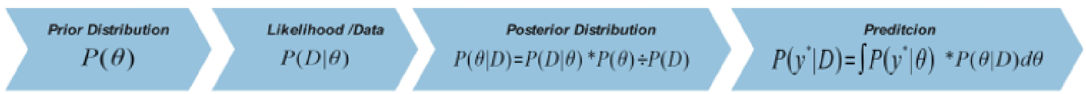


Figure 3: process structure

According to the above ideas, we can estimate the country that won the medal for the first time, and Model predictions for countries that have not yet won medals for further evaluation Which countries are expected to win medals for the first time in 2028. According to the data, since each country has won medals in history, we consider Countries without medals in 2024 but expected to win medals in 2028 are screened out. The results are as follows: The following table(Show only part of the data).

Table 4: Predicted First-Time Medal Winning Countries

NOC	Number of medals	Average Winning Probability
Zambia	1	0.602
Taiwan	2	0.659
Virgin Islands	1	0.595
British West Indies	1	0.559
Zambia	1	0.658

4.3.2 Defining Similarity Based on Event Characteristics and Historical Data

The core of the Bayesian hierarchical model lies in constructing multilevel statistical models to handle data with nested or multilevel structures. For the five new events added to the 2028 Los Angeles Olympics in the United States—baseball, softball, cricket,flag football, and lacrosse—although we lack direct data, we can predict the potential medal counts for these new events across different countries by leveraging the similarities with other sports.

To predict the award outcomes for these new events, we compare them with other existing Olympic events based on the following qualitative characteristics:

1. **Step 1** Selection of Similar Indicators
   - Event Category and Nature:
   - Competition Rules and Organization:

- Popularity Among Participating Countries:
- Athletic Requirements of the Event:
- Competition Venue and Equipment:

2. **Step 2** Selection of Similar Projects Based on Similar Indicators:
Through the aforementioned analysis, we have obtained indicators for measuring the similarity between different events. Next, by conducting a qualitative comparative analysis between the new event and existing events, we can identify the similar events we need. Leveraging the Bayesian hierarchical model and studying these similar events, we can predict the number of medals that different countries are likely to win in the new event.

3. **Step3** Prediction of Award-Winning Status for New Projects Based on Bayesian Models:
After comparing the new projects with existing ones based on similar indicators, we have identified "similar projects" that highly match each of the new projects. Next, we will use the historical award-winning situations (gold, silver, and bronze medals) of these similar projects as direct references and employ a Bayesian hierarchical model to make predictions for the new projects. Due to the limited amount of data we have, using a Bayesian hierarchical modeling approach and prior distributions may lead to overfitting, which would result in significant errors in the prediction results. To avoid this, we will consider a direct mapping approach, which involves directly "filling in" the medal distribution for the new projects using the medal distribution of the similar projects. This includes:After comparing the new projects with existing ones based on similar indicators, we have identified "similar projects" that highly match each of the new projects. Next, we will use the historical award-winning situations (gold, silver, and bronze medals) of these similar projects as direct references and employ a Bayesian hierarchical model to make predictions for the new projects. Due to the limited amount of data we have, using a Bayesian hierarchical modeling approach and prior distributions may lead to overfitting, which would result in significant errors in the prediction results. To avoid this, we will consider a direct mapping approach, which involves directly "filling in" the medal distribution for the new projects using the medal distribution of the similar projects. This includes:

   - **Simple Weighted Average** :
   Assuming the number of gold, silver, and bronze medals for similar projects are $G_1, G_2, \ldots, G_n$, $S_1, S_2, \ldots, S_n$, and $B_1, B_2, \ldots, B_n$ respectively, we can directly calculate their means and standard deviations as follows:
   $$\mu_{\text{gold}} = \frac{1}{n}\sum_{i=1}^{n} G_i, \mu_{silver} = \frac{1}{n}\sum_{i=1}^{n} S_i, \mu_{bronze} = \frac{1}{n}\sum_{i=1}^{n} B_i$$
   - **Statistical Summary:** These means can be used to represent the predicted number of gold, silver, and bronze medals for newly added projects.

4.4 Prediction of Award-Winning Probability for Future Countries Based on Socioeconomic Factors

Regarding the prediction of the award-winning probability for countries that have never won an award at the 2028 Los Angeles Olympics, as addressed in Question 2, we believe that this problem still highly aligns with the application scope of Bayesian hierarchical models. By analyzing the similarities between countries that have never won an award and those that have, we can screen out the "similar countries" that have won awards, which are relevant to our needs. By assessing the award-winning probability of these similar countries in the next Olympic Games, we can measure the probability of countries that have never won an award to win an award for the first time at the 2028 Olympics. This approach will yield results that meet our expectations.

- Step 1: Selection of Similarity Indicators
  Per Capita GDP:
  Population Size:
  Cultural and Sporting Participation:
  Climate and Geographic Environment:

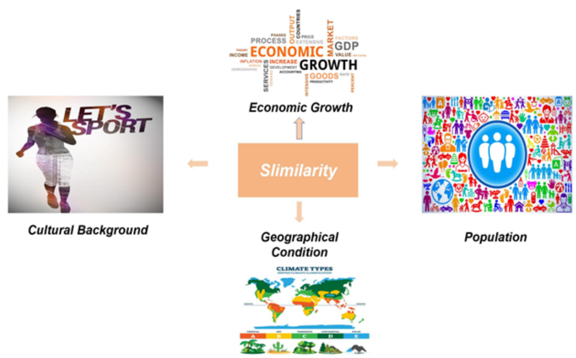


Figure 4: dicators

- Step 2: Screening Similar Countries Based on Similarity Indicators
  In Step 1, we have already obtained the similarity indicators for selecting similar countries. Next, we can use these indicators to gradually screen out countries from those that have won awards, which highly match the similarity indicators of countries that have never won an award.

- Step 3: Predicting the Award-Winning Probability for Non-Award-Winning Countries in the Next Olympic Games Based on a Bayesian Model

  1. Data Collection and Statistics:
     We assume that we have data from multiple similar countries, including the number of gold, silver, and bronze medals won by each country in both 2020 and 2024.

     Our prediction is divided into two major modules: In the first step, we will predict the probability of a similar country winning gold, silver, and bronze medals in new events based on the proportion of these medals among the total medals won by that similar country. In the second step, we will use the average number of gold, silver, and bronze medals won by similar countries in 2020 and 2024 as a direct reference to estimate the award-winning situation for non-award-winning countries in the next Olympic Games, including both existing and new events. Therefore, our modeling steps can be divided as follows:

Gold Medals/Silver Medals/Bronze Medals: The country won G/S/B2020 times in 2020 and G/S/B2024 times in 2024.

2. Calculating the Average Number of Each Medal Type:
Firstly, we compute the average number of each medal type (gold, silver, and bronze) won by all similar countries in both 2020 and 2024. Subsequently, we use these averages as the benchmarks for predicting the performance of new events.

$$\mu_{\text{gold/silver/bronze}} = \frac{1}{N_{total}} \sum_{i=1}^{N_{countries}} (\sum_{j=2}^{2} G_{year,i,j})$$

Where:

N_{total} : denotes the total number of similar countries

$G_{\text{year},i,j}$: represents the number of gold medals won by the ith country in the jth year of a specific per

3. Calculate the Medal Percentage for Each Country:
In order to reflect the historical performance ratio of each similar country, we calculate the percentage of gold, silver, and bronze medals it has won out of the total number of medals awarded in that year:

$$gold/silver/bronze\_\text{ratio}_i = \frac{\sum_{j=1}^{2} G_{year,i,j}}{\sum_{i=1}^{N_{countries}} G_{year,i,j}}$$

4. Predict the number of gold, silver, and bronze medals for new events among similar countries.
Unlike the direct prediction of medal awards for new events in the previous question, this subsection focuses on predicting the number of gold, silver, and bronze medals for new events at the national level. However, the underlying principle remains the same: we need to predict the number of gold, silver, and bronze medals for new events based on the historical performance ratios of similar countries in existing events. The specific approach is outlined below:

$$\overset{\Lambda}{G}/\overset{\Lambda}{S}/\overset{\Lambda}{B} = \mu_{gold/silver/bronze} \times gold/silver/bronze\_ratio_i$$

Where G^{Λ} represents the predicted quantity of gold medals, S^{Λ} represents the predicted quantity of silver medals, and B^{Λ} represents the predicted quantity of bronze medals.

5. Calculate the Probability of Winning Each Award:
Based on the predictions of the number of gold, silver, and bronze medals, similarly, we can obtain the corresponding probability of winning each award. Assuming the number of medals obtained by a country is:

$$\overset{\Lambda}{T} = \overset{\Lambda}{G} + \overset{\Lambda}{S} + \overset{\Lambda}{B}$$

$$P(gold, (silver, bronze)) = \frac{\overset{\Delta}{G}(\overset{\Delta}{S}, \overset{\Delta}{B})}{\overset{\Delta}{T}}$$

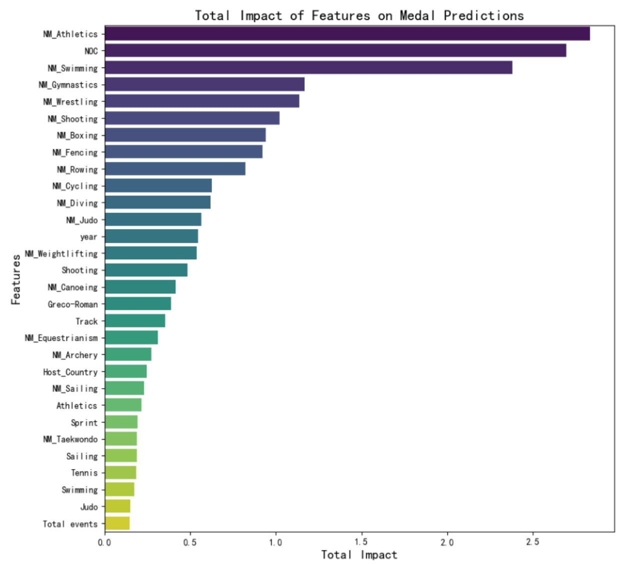


Figure 5: Impact of events on the number of medals

For the projects selected by the host country, they are usually influenced by the support and excellent resources of the host country, which will have a gain effect

5 Task 2:Quantitative analysis of the "Great Coach" effect

5.1 Data preprocessing

In the quantitative analysis of the "great coach" effect, data preprocessing is a crucial step that provides accurate and effective support for subsequent analysis. To ensure the reliability of the analysis results, we will focus on organizing the performance data under the leadership of Coach Liu Guoliang (coach of the Chinese men's table tennis team), Coach Li Jun (coach of the Chinese women's table tennis team), and Coach Li Yongbo in the badminton field in China. The specific data preprocessing steps are as follows:

1. Data Selection: From the "Summer Olympic Athletes Dataset," we select the performance data of the Chinese men's table tennis and badminton teams from 1988 to 2016. The selection criteria are as follows: male athletes in the table tennis event coached by Liu Guoliang, male athletes in the badminton event coached by Li Yongbo, with the players being from China and the year being before 2016. Through these selection criteria, we are able to obtain the medal data of the three coaches during their coaching period, providing a reliable data foundation for subsequent analysis.

2. Medal Score Calculation: To conveniently assess the performance of each athlete, we convert the medal types into numerical values for statistical analysis. The mapping rules for the medals are as follows:

$$\text{Medal\_score} = \begin{cases} 0, & \text{No medal} \\ 1, & \text{Bronze} \\ 2, & \text{Silver} \\ 3, & \text{Gold} \end{cases}$$

Using this mapping rule, the medal types are quantified into numerical values. Then, for each athlete, based on the year group, we calculate the total score S for each year's medals as follows:

$$S = \sum_{i=1}^{n} \text{Medal\_score}_i$$

Where n represents the number of athletes in that year, and n is the total number of athletes participating in the competition that year. This way, we can obtain the total medal score for "Chinese men's table tennis at the Olympics each year."

In order to analyze the "great coach" effect, we need to construct a new variable, "Is Coaching," represented by Q. This variable indicates whether the coaching occurred before or after the coach started their tenure, and it is defined as follows:

$$Q = \begin{cases} 0, & \text{Year} < \text{Year coach started} \\ 1, & \text{Year} \geq \text{Year coach started} \end{cases}$$

3. By defining this variable, the data can be divided into two groups: before the coach started their tenure and after they started, to provide support for subsequent hypothesis testing. The coaching situation for the three coaches is shown as follows:

Table 5: Coaching Data for Liu Guoliang, Li Jun, and Li Yongbo

	Liu Guoliang		**Li Jun**		**Li Yongbo**	
Year	**S**	**Q**	**S**	**Q**	**S**	**Q**
≤ 1988	0	0	0	0	0	0
1992	1	0	5	0	2	0
1996	5	0	4	1	4	1
2000	4	0	3	1	12	1
2004	3	1	3	1	8	1
2008	15	1	15	1	9	1
2012	14	1	14	1	9	1
2016	14	1	14	1	3	1

Through the above steps, the dataset was processed into tables about Year, S, and Q, providing a reliable data foundation for subsequent hypothesis testing, ensuring data consistency and reproducibility, and laying a solid foundation for the accuracy and effectiveness of quantitative analysis of the "Great Coach" effect.

5.2 Modeling

In this step, the hypothesis testing method is used to evaluate the effect of the "Great Coach" by matching the sample t-test. The hypothesis is tested by evaluating the effect of the coach on the total number of medals won by the Chinese men's table tennis team. Three types of coaching situations are considered, and the hypothesis is tested to determine whether the change in the number of medals after the coach's appointment is significant.

• Model Hypothesis

The original hypothesis H_0: Liu Guoliang as the coach of the Chinese men's table tennis team has no significant effect on the change in the total number of medals won by the team.

$$H_0 : \mu_{\text{pre}} = \mu_{\text{post}}$$

Where μ_{pre} represents the mean number of medals won before coaching, and μ_{post} represents the mean number of medals won after coaching.

The alternative hypothesis H_1: After Liu Guoliang became the coach, there was a significant change in the total number of medals won by the Chinese men's table tennis team, which indicates that coaching had a significant effect.

$$H_1 : \mu_{\text{pre}} \neq \mu_{\text{post}}$$

• t-Test for Matching Samples

The t-test for matched samples is used to test whether there is a significant difference between the means of two related samples. In this analysis, we compare the data before and after Liu Guoliang became the coach and perform a t-test on this matched sample.

The formula for the t-test is as follows:

$$t = \frac{d}{s_d/\sqrt{n}}$$

Where d represents the difference between the sample means, s_d is the standard deviation of the differences, and n is the number of paired samples. The difference d_i represents the difference between the number of medals won before and after coaching.

1. Difference Calculation:

For each pair of data, calculate the difference in the total number of medals won before and after coaching:

$$d_i = d_{\text{post}} - d_{\text{pre}}$$

2. Hypothesis Testing:

Calculate the mean and standard deviation of the differences; calculate the t-value, and check the distribution using the degrees of freedom $n-1$. Based on the t-statistic and p-value, decide whether to reject the null hypothesis.

3. Conclusion of Hypothesis:

Based on the result of the hypothesis test at the significance level α (commonly taken as 0.05), if the p-value is less than the significance level, then reject the null hypothesis and conclude that the "Great Coach" effect exists. That is:

If $p < 0.05$, reject the null hypothesis and conclude that the "Great Coach" effect exists;

If $p > 0.05$, fail to reject the null hypothesis and conclude that the "Great Coach" effect does not exist.

Using the matched sample t-test, we confirm that the statistical data and p-value. If the p-value is less than the significance level $\alpha = 0.05$, we can conclude that Liu Guoliang's coaching has led to a significant change in the total number of medals won by the Chinese men's basketball team.

If the p-value is greater than 0.05, we cannot reject the null hypothesis and conclude that Liu Guoliang's coaching did not significantly affect the total number of medals won by the team.

5.3 Model Solution

1. Calculation of the Difference:

For each pair of data, calculate the difference in the total number of medals won before and after coaching (d_i):

$$d_i = d_{\text{post}} - d_{\text{pre}}$$

For example:

$$d_1 = d_{2004} - d_{1998} = 3 - 0 = 3$$

Step 2: Calculate the Mean and Standard Deviation of the Differences:

The mean value of the differences $\overline{d}$ is calculated as follows:

$$\overline{d} = \frac{\sum_i d_i}{n} = \frac{3 + 14 + 9 + 10}{4} = 9$$

Step 3: Calculate the t-Statistic:

The formula for the t-statistic is:

$$t = \frac{\overline{d}}{s_d/\sqrt{n}}$$

Where $n = 4$ (the number of samples) and $\overline{d} = 9$, and $s_d = 4.54$.

Substituting in the values:

$$t = \frac{9}{\frac{4.54}{\sqrt{4}}} \approx \frac{9}{2.27} \approx 3.96$$

Step 4: Calculate the p-Value and Make a Decision:

Based on the degrees of freedom $F = n - 1 = 3$, we look up the corresponding p-value in the t-distribution table. The calculated p-value is:

$$p = 0.0288$$

Step 5: Hypothesis Test Results:

Given that the significance level is $\alpha = 0.05$, since $p = 0.0288$ is less than 0.05, we reject the null hypothesis.

This implies that the coaching of Liu Guoliang had a significant effect on the total number of medals won by the Chinese men's table tennis team, and thus supports the existence of the "Great Coach" effect.

5.4 The impact of the 'Great Coach' effect on medal count

Through regression analysis, we quantify the "Great Coach" effect on the total number of medals won by the Chinese men's tennis team. Taking Liu Guoliang as an example, we use the least squares method to evaluate the effect of coaching changes on the total number of medals. The regression model is as follows:

$$\text{Medal\_score}_i = \beta_0 + \beta_1 \times Q + \epsilon_i$$

In the regression analysis, *Medal_score* represents the number of medals won in year i, Q is the binary variable indicating whether there was a coaching change, β_0 is the intercept term, β_1 is the regression coefficient that indicates the effect of the coaching change on the number of medals, and ϵ_i is the error term.

The regression analysis estimates β_1, and through statistical tests, we can determine its significance. If the P-value is less than 0.05, it indicates that the coaching change had a significant impact on the number of medals won; if the P-value is greater than 0.05, it indicates that the effect is not significant.

The results of the regression show that β_1 is positive and significant, indicating that Liu Guoliang's coaching had a positive impact on the number of medals won, providing evidence of the "Great Coach" effect.

Based on the results of the OLS regression analysis, the model quantifies the effect of the "Great Coach" on the total number of medals won by the Chinese men's table tennis team. By analyzing

Liu Guoliang's coaching and its relationship with the number of medals won, we obtain the following regression results and statistics from other coaching scenarios:

Table 6: OLS Regression Results for Liu Guoliang, Li Jun, and Li Yongbo (n=8)

Liu Guoliang OLS Regression Results (n=8)					
Coef	**Std.err**	**t**	**P**	**VIF**	**R2**
Constant	2.5	2.179	1.147	0.295	-
Q	9.0	3.082	2.920	0.027**	0.787
Li Jun OLS Regression Results (n=8)					
Coef	**Std.err**	**t**	**P**	**VIF**	**R2**
Constant	4.2	3.9	1.126	0.213	-
Q	7.6	3.2	2.813	0.017**	0.722
Li Yongbo OLS Regression Results (n=8)					
Coef	**Std.err**	**t**	**P**	**VIF**	**R2**
Constant	3.0	2.12	1.417	0.218	-
Q	5.2	2.076	2.057	0.042**	0.716

Note: *, **, *** represent significance at the 1%, 5%, and 10% levels, respectively.

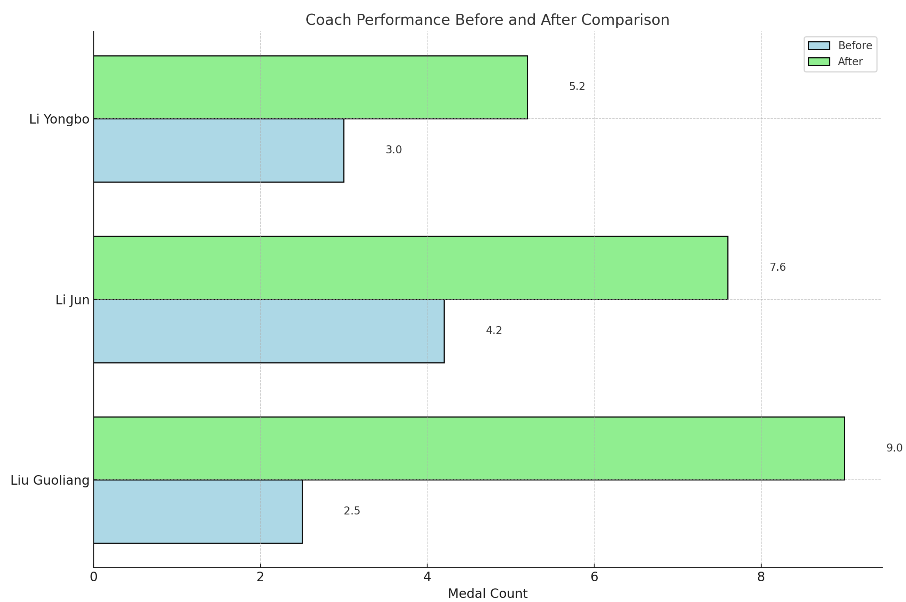


Figure 6: Comparison before and after coaching

From the regression results, we can extract the following key information: the correlation coefficient R^2 is generally greater than 0.7, which shows that about 70

Analysis shows that teacher-led instruction improves the reward points of students, and the regression coefficient for the number of instructional hours provided by the teacher has a supporting role. This suggests that other countries that selected top-tier teaching professionals based on the data, through investing in instructional resources, optimized their strategic layout.

5.5 Strategic recommendations for coach investment

5.5.1 Investment priorities for coaches in various countries

On a global scale, optimizing national investment in sports is a key step to improve national sports performance. In order to accurately assess which countries and projects are worth investing in, the experience of coaches and the effect of investment must be fully considered. First of all, we need to analyze the participation of athletes from various countries, such as the data of badminton, to assess which countries have performed well in this field and have great development potential. We have made statistics on the number of athletes participating in badminton competitions every year, and combined with the achievements in the past few years, made in-depth analysis from multiple angles, and provided suggestions for the direction of investment based on these data.

To obtain the most representative data, we summarize the participation status of each country and perform deeper analysis on their sports data, including calculating the average number of participants for each country:

$$\text{Average Participants} = \frac{1}{n}\sum_{i=1}^{n}\text{Count}_{ij}$$

Where Count represents the number of participants in each country for a specific sports event in that year, and the total number for that event. By selecting the countries with the highest participation numbers, we can better understand where these countries have more potential and provide them with more investment recommendations.

The following table shows examples of badminton events in the top three countries based on the calculation results and the number of participants:

Table 7: Top 3 Countries with the Most Participants in Badminton

Team	Count
Korea	13
China	8.7
Denmark	7.7

The participation rate of badminton events in these countries is the highest, indicating that their sports activities and participation level are relatively high. Therefore, the investment suggestions for these countries will focus more on strengthening their badminton events and further help them improve their overall ranking.

5.5.2 Prediction of return on coach investment

To quantify the effect of coach investment, this study will use regression analysis based on the previous framework, where the investment in coaching resources across different countries will be used as the reference for future predictions. By analyzing the performance of various countries in badminton, the projected impact of investing in “excellent” coaching on the increase in medal counts will be predicted.

To analyze the impact of coach investment on the performance of different countries in badminton, the study first conducts a regression analysis, calculating the effect of coach Li Yongbo’s investment. The coefficient ($\beta_3 = 5.2$) demonstrates the effect of Li’s coaching on China’s badminton team. This value represents the average increase of 5.2 medals in the number of medals won by the team. Based on this model, the combined effect of coach investment, along with other countries’ average medal counts, is used to predict the average medal count for each country after investing in “excellent” coaching.

(1) **Standard Medal Count Calculation** By calculating the total number of awards each country wins each year, we can compute the average number of awards for each country, μ_{country}, which is given by the formula:

$$\mu_{\text{country}} = \frac{1}{n}\sum_{i=1}^{n} \text{Medal score}_i$$

Where, Medal score$_i$ represents the number of medals in year i, and n is the total number of years for that country.

(2) **Adjustment Factor Calculation** Based on data from section 5.1.1, during the pre-training period for Li Yongbo’s national team, the average number of medals for China’s Badminton team is $\mu_{\text{China}} = 3.0$. The adjustment factor can be calculated by comparing the average medals of each country with that of China, as shown in the formula below:

$$\text{adjustment factor} = \frac{\mu_{\text{country}}}{\mu_{\text{China}}}$$

This adjustment factor reflects how the awards for each country compare to China’s performance under the same conditions. A higher adjustment factor indicates that the country has more competitive potential.

(3) **Predicted Medal Growth** Using the adjustment factor and Li Yongbo’s effective value β_3, the predicted medal growth for each country after investing in teaching is calculated as follows:

$$\text{Predicted Growth}_i = \text{adjustment factor}_i \times \beta_3$$

Where Predicted Growth represents the predicted increase in medals for country i after investing in "excellent" coaching.

(4) **Predicted Medal Count** Finally, the predicted growth in medals is added to each country’s base medal count to obtain the predicted total number of medals. The formula is:

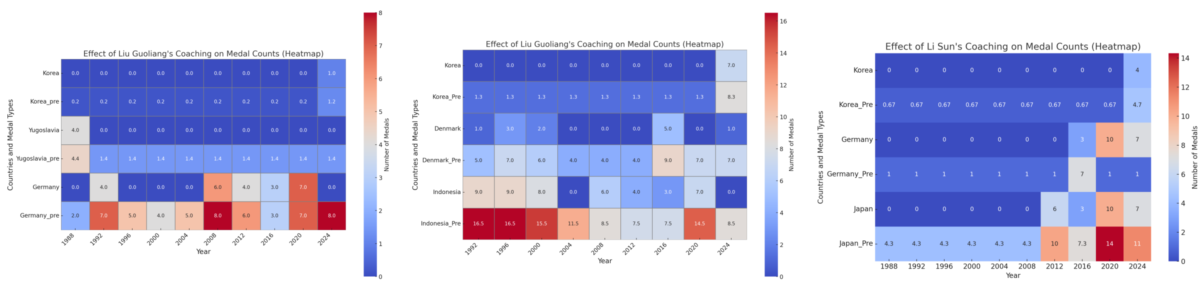
$$\text{Predicted Medal}_i = \mu_{\text{country}} + \text{Predicted Growth}_i$$

This value represents the predicted total number of medals for country i after investing in teaching, based on the hypothetical performance in the context of Li Yongbo’s national team. According to these calculations, the following predicted results were obtained:

Table 8: Predicted Medal Counts for Korea, Denmark, and Indonesia

Year	Korea	Denmark	Indonesia	Korea Pre	Denmark Pre	Indonesia Pre
1992	0.0	1.0	9.0	1.4	5.0	16.5
1996	0.0	3.0	9.0	1.4	7.0	16.5
2000	0.0	2.0	8.0	1.4	6.0	15.5
2004	0.0	0.0	4.0	1.4	4.0	11.5
2008	0.0	1.0	4.0	1.4	4.0	8.5
2012	0.0	4.0	3.0	1.4	8.0	7.5
2016	0.0	5.0	0.0	1.4	9.0	7.5
2020	0.0	3.0	7.0	1.4	7.0	14.5
2024	7.0	3.0	1.0	8.4	7.0	8.5

Similarly, based on the assumption of different coaching effects, the predicted results are obtained through the method outlined above. Below, heatmaps are created for the three assumed coaching effects to represent the changes in the medal counts of each country.



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(a) (b) (c)

Figure 7: Heat map of coaching effect

Based on the above results, the following strategic recommendations are derived from the three selected countries:

•**India**: By investing in high-quality coaching, improve competitive levels and scientific training capabilities, combine sports science and technology analysis, and improve the competitive status. Strengthen coaching facilities, promote the application of technological innovation and new strategies, and improve overall competitiveness.

•**Denmark**: Denmark's medal predictions in badminton are optimistic. To enhance its competitiveness, Denmark can focus on developing young athletes by improving their fundamental techniques and tactical awareness. Additionally, promoting international cooperation and inviting top coaches to train local coaches can further improve the level of domestic athletes. Through these efforts, Denmark is expected to strengthen its international competitiveness.

•**Japan**: Continue to invest in technical innovation, focus on detailed technical training to achieve success, especially in global and Olympic events. Strengthen youth systems, invest in advanced training facilities, and increase international competition exposure, improving athlete selection for competitions.

In conclusion, through the prediction of coaching investments, countries can receive tailored coaching investment recommendations that help evaluate their country-specific models.

6 Model Insights and Discussion

6.1 Innovation point and Weaknesses

This study uses the multivariate negative binomial distribution and Bayesian hierarchical models to predict medal counts, combined with hypothesis testing and regression models to assess the "great coach" effect. The model incorporates factors such as athlete differences, national competitiveness, host country effects, and social media data, improving prediction accuracy. By dynamically adjusting model parameters, the Olympic Committee can optimize preparation strategies. Additionally, a new variable is constructed, and the hypothesis testing method is used to quantify the "great coach" effect, providing the Olympic Committee with a method to test and quantify the "great coach" effect.

While the multivariate negative binomial and Bayesian hierarchical models effectively capture the data structure and improve prediction accuracy, they face challenges due to their high computational complexity, sensitivity to data quality, and reliance on computationally intensive MCMC methods. These factors make the model difficult to apply in practical scenarios. The sample size for the models used in hypothesis testing and regression analysis is relatively small, which may affect the statistical significance and reliability of the results. Furthermore, the coach's effect is influenced by various factors, and evaluating one coach may not be universally applicable. Additionally, the model does not sufficiently account for external factors such as athletes' individual abilities, training conditions, and international competition experience, all of which may significantly impact medal counts.

6.2 Sensitivity Analysis

The improved Bayesian hierarchical model exhibits strong stability, as evidenced by minimal changes in prediction results when different datasets and hypotheses are included. Adding additional data points and considering factors specific to different countries can result in minor adjustments to model parameters. These small changes indicate that the model can remain stable under different assumptions and capture the relationship between countries, sports events, and historical performance.

In hypothesis testing and regression analysis, sample size is a key factor. For example, when analyzing the impact of Liu Guoliang's coaching, the lack of data points before and after his coaching resulted in unstable prediction results. However, the model remains robust in identifying trends and positive relationships, especially when identifying potential effects of 'great coaches', despite the limited sample size. This model demonstrates strong stability towards key predictive variables such as coaching effects and Olympic success potential based on historical performance. However, they are also sensitive to specific assumptions such as data availability and the influence of external factors such as athlete training and international experience.

6.3 Possible Improvements

Future improvements could involve combining these models with dynamic updating methods and incorporating more granular data, such as athlete-level performance. Additionally, integrating machine learning techniques could enhance prediction accuracy and adaptability, making these models more useful for the Olympic Committee's decision-making processes.

References

[1] Ramchandani, M. G., Wilson, et al. Review of medal predictions for South Africa in the Delhi 2010 Commonwealth Games: commentary. *South African Journal of Sports Medicine*, 2010, 22(4): 106-107.

[2] Science - Social Science; Study Findings from Manchester Metropolitan University Provide New Insights into Social Science (Forecasting National Medal Totals At the Summer Olympic Games Reconsidered). *Science Letter*, 2020.

[3] Sanchez-Fernandez, Patricio, Vaammonde-Liste, et al. Olympic medals: success predictions for Rio-2016. *South African Journal for Research in Sport, Physical Education and Recreation*, 2016, 38(3): 195-206.

[4] Vagenas, G., Vlachokyriakou, E. Olympic medals and demo-economic factors: Novel predictors, the ex-host effect, the exact role of team size, and the "population-GDP" model revisited. *Sport Management Review*, 2012, 15(2): 211-217.